

INTRODUCTION:

Artificial intelligence (AI) is the capability of [computational systems](#) to perform tasks typically associated with [human intelligence](#), such as [learning](#), [reasoning](#), [problem-solving](#), [perception](#), and [decision-making](#). It is a [field of research](#) in [engineering](#), [mathematics](#) and [computer science](#) that develops and studies methods and [software](#) that enable machines to perceive their environment and use [learning](#) and [intelligence](#) to take actions that maximize their chances of achieving defined goals.^[1]

High-profile [applications of AI](#) include advanced [web search engines](#), [chatbots](#), [virtual assistants](#), [autonomous vehicles](#), and play and analysis in [strategy games](#) (e.g., [chess](#) and [Go](#)). Since the 2020s, [generative AI](#) has become widely available to generate images, audio, and videos from text prompts.

The traditional goals of AI research include learning, [reasoning](#), [knowledge representation](#), [planning](#), [natural language processing](#), and [perception](#), as well as support for [robotics](#).^[a] To reach these goals, AI researchers have used techniques including [state space search](#) and [mathematical optimization](#), [formal logic](#), [artificial neural networks](#), and methods based on [statistics](#), [operations research](#), and [economics](#).^[b] AI also draws upon [psychology](#), [linguistics](#), [philosophy](#), [neuroscience](#), and other fields.^[2] Some companies, such as [OpenAI](#), [Google DeepMind](#) and [Meta](#), aim to create [artificial general intelligence](#) (AGI) – AI that can complete virtually any cognitive task at least as well as a human.^[3]

Artificial intelligence was founded as an academic discipline in 1956,^[4] and the field went through multiple cycles of optimism throughout [its history](#).^{[5][6]} followed by periods of disappointment and loss of funding, known as [AI winters](#).^{[7][8]} Funding and interest increased substantially after 2012, when [graphics processing units](#) began being used to accelerate neural networks, and [deep learning](#) outperformed previous AI techniques.^[9] This growth accelerated further after 2017 with the [transformer architecture](#).^[10] In the 2020s, an [AI boom](#) has coincided with advances in [generative AI](#), which allowed for the creation and modification of media. In addition to [AI safety](#) and [unintended consequences and harms](#) from the use of AI, [ethical concerns](#), [AI's long-term effects](#), and [potential existential risks](#) have prompted discussions of [AI regulation](#).

Natural Language Processing:

[Natural language processing](#) (NLP) allows programs to read, write and communicate in human languages.^[50] Specific problems include [speech recognition](#), [speech synthesis](#), [machine translation](#), [information extraction](#), [information retrieval](#) and [question answering](#).^[51]

Early work, based on [Noam Chomsky's generative grammar](#) and [semantic networks](#), had difficulty with [word-sense disambiguation](#)^[f] unless restricted to small domains called "[micro-worlds](#)" (due to the [common sense knowledge problem](#)^[28]). [Margaret Masterman](#) believed that it was meaning and not grammar that was the key to understanding languages, and that [thesauri](#) and not dictionaries should be the basis of computational language structure.

Modern deep learning techniques for NLP include [word embedding](#) (representing words, typically as [vectors](#) encoding their meaning),^[52] [transformers](#) (a deep learning architecture using an [attention](#) mechanism),^[53] and others.^[54] In 2019, [generative pre-trained transformer](#) (or "GPT") language models began to generate coherent text,^{[55][56]} and by 2023, these models were able to get human-level scores on the [bar exam](#), [SAT](#) test, [GRE](#) test, and many other real-world applications.^[57]

TECHNIQUES:

State space search:

[State space search](#) searches through a tree of possible states to try to find a goal state.^[69] For example, [planning](#) algorithms search through trees of goals and subgoals, attempting to find a path to a target goal, a process called [means-ends analysis](#).^[70]

[Simple exhaustive searches](#)^[71] are rarely sufficient for most real-world problems: the [search space](#) (the number of places to search) quickly grows to [astronomical numbers](#). The result is a search that is [too slow](#) or never completes.^[13] "[Heuristics](#)" or "rules of thumb" can help prioritize choices that are more likely to reach a goal.^[72]

[Adversarial search](#) is used for [game-playing](#) programs, such as chess or Go. It searches through a [tree](#) of possible moves and countermoves, looking for a winning position.^[73]

Local search:

[Local search](#) uses [mathematical optimization](#) to find a solution to a problem. It begins with some form of guess and refines it incrementally.^[74]

[Gradient descent](#) is a type of local search that optimizes a set of numerical parameters by incrementally adjusting them to minimize a [loss function](#). Variants of gradient descent are commonly used to train [neural networks](#),^[75] through the [backpropagation](#) algorithm.

Another type of local search is [evolutionary computation](#), which aims to iteratively improve a set of candidate solutions by "mutating" and "recombining" them, [selecting](#) only the fittest to survive each generation.^[76]

Distributed search processes can coordinate via [swarm intelligence](#) algorithms. Two popular swarm algorithms used in search are [particle swarm optimization](#) (inspired by bird [flocking](#)) and [ant colony optimization](#) (inspired by [ant trails](#)).^[77]

Deep learning:

[Deep learning](#) uses several layers of neurons between the network's inputs and outputs.^[109] The multiple layers can progressively extract higher-level features from the raw input. For example, in [image processing](#), lower layers may identify edges, while higher layers may identify the concepts relevant to a human such as digits, letters, or faces.^[113]

Deep learning has profoundly improved the performance of programs in many important subfields of artificial intelligence, including [computer vision](#), [speech recognition](#), [natural language processing](#), [image classification](#),^[114] and others. The reason that deep learning performs so well in so many applications is not known as of 2021.^[115] The sudden success of deep learning in 2012–2015 did not occur because of some new discovery or theoretical breakthrough (deep neural networks and backpropagation had been described by many people, as far back as the

1950s)^[i] but because of two factors: the increase in computer power (including the hundred-fold increase in speed by switching to [GPUs](#)) and the availability of vast amounts of training data, especially the giant [curated datasets](#) used for benchmark testing, such as [ImageNet](#).^[j]